Wadim **Müller**

Embedded Software Expert · Linux Enthusiast

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"He who stops being better stops being good." (Oliver Cromwell)

Education

Technische Hochschule Aschaffenburg (University of Applied Sciences)

Aschaffenburg, Germany

B.ENG. IN ELECTRICAL ENGINEERING AND INFORMATION TECHNOLOGY — *Sep. 2008 - Aug. 2012*

- Emphasis on Microelectronics with final grade of 1.5 (First Class Honours).
- Thesis: Development of an USB Device Interface based on the Test and Measurement USB Class Standard (USB TMC).

Hochschule Darmstadt (University of Applied Sciences)

Darmstadt, Germany

M.SC. IN ELECTRICAL ENGINEERING — *Sep. 2014 - Aug. 2017*

- Extra-occupational study with emphasis on Microelectronics with final grade of 1.1 (First Class Honours).
- Thesis: Development of an FPGA accelarated Linux System for Realtime Video Feature extraction using OpenCV on a Zynq Multicore pSoC. https://github.com/wafgo/morphological-filter-ip-core, https://github.com/wafgo/disparity-coprocessor-ip-core, https://github.com/wafgo/linux-kernel-custom-hw-uio, https://github.com/wafgo/rt-depth-map.

Skills

Programming	C, C++, Rust, Python, Perl, Bash
Kernel & Low-Level	Linux kernel drivers (IIO, counter, PWM, GPIO, I²C, PCI/PCI-EP), devicetree, U-Boot, TF-A, remoteproc/RPMsg, FreeRTOS, AU Classic
Architectures	ARMv7/v8-A, Cortex-M/R, RISC-V, Renesas SH2A, NXP S12Z, Xilinx Zynq pSoC
Virtualization/Emulation	QEMU (SoC, machine and device models), KVM
Debugging	Lauterbach Trace32, GDB, OpenOCD, Greenhills MULTI, ARM CoreSight/ETM, JTAG, KUnit
Build & Distribution	Yocto/OpenEmbedded (bitbake), CMake, GNU make, kbuild, gcc, binutils, llvm/clang, ghs, capstone, GitLab CI
Scientific Computing	PySpark/Databricks, RDKit, GFN2-xTB/CREST and DFTB+ (semi-empirical), Psi4 (DFT)
Languages	German, English, Russian

Open Source Contributions

Linux Kernel (mainline)

drivers/iio/flow/slf3s.c

Maintainer · Sensirion SLF3S Flow Sensor Driver — *since 2026*

- Author and listed maintainer of the upstream IIO driver for the SLF3S liquid flow sensor, merged for v7.3, including the devicetree binding, the required additions to the IIO core and KUnit test coverage.

Linux Kernel (mainline)

lore.kernel.org

Contributor · SoC Enablement, Drivers and Tooling — *since 2022*

- Upstreamed the initial mainline bring-up of the **NXP S32G3** automotive SoC: board devicetree plus the `linflexuart` and `esdhc` bindings. Series (v4).
- Added **RISC-V support to `scripts/checkstack.pl`**, so worst-case stack usage can be analysed on that architecture. Patch.
- Authored a generic **GPIO-based counter driver**, currently under upstream review. Series (v7).
- Proposed a **PCI Endpoint block-device passthrough** function driver and its host-side counterpart, allowing block devices to be shared between a PCI host and endpoint. Series, covered on LWN.

QEMU

lore.kernel.org/qemu-devel

Contributor · TI AM64x SoC and Machine Model — *since 2026*

- Contributed a **TI AM64x (Sitara AM6442) SoC model and `am64-virt` machine**, complete enough to run the stock TI boot chain end to end: ROM boot, R5 SPL, TF-A/OP-TEE, U-Boot and SMP Linux. Includes models for the DMSC (TI-SCI) system controller, secure proxy, TI mailbox, RAT address translation, K3 DMTimer and a parser for TI's combined boot image. Series (RFC v2, 14 patches).
- Fixed a generic **SDHCI ADMA descriptor-chain bug** and contributed the qtest reproducer for it. Fix.
- Earlier: machine emulation of customised Renesas R-Car3 boards, booting a proprietary RTOS from the reset vector (github.com/wafgo/qemu), and a model of the NXP S32G Cortex-M7 cluster running an unmodified Classic AUTOSAR stack.

Various Projects

github.com/wafgo

Contributor · Debuggers, Binary Tooling and Quantum Chemistry — *since 2018*

- **pyelftools**: added the V850/RH850 relocation enum, which is what lets DWARF and relocation data of Renesas RH850 objects be parsed (merged).
- **Psi4**: proposed GauXC as an optional backend for the exchange-correlation integration in this quantum-chemistry package (PR).
- **GEF** (GDB extension): RISC-V support, including branch instruction decoding (merged); smaller fixes in **radare2**, **pahole/dwarves** and **ccls**.
- Own projects, among them WrapMaster (dynamic function wrapping for debugging), a R-Car3 boot ROM reimplementation, Trace32 scripts generating the R-Car3 DBSC4 DDR4 controller init, and an SDK for the Grain Media GM8136 SoC.

Experience

CMBlu Energy AG

Alzenau, Germany

Embedded Linux Developer · Linux Distribution Maintainer — *since Feb. 2025*

- Own and maintain the company's in-house Yocto-based Linux distribution for our TI AM64x platform (Cortex-A53 plus a Cortex-M4 safety core): BSP and kernel integration, custom layers and recipes, image and RPM feed, and the over-the-air update path for the deployed fleet.
- Performed complete board bring-up and hardware validation for custom AM64xx PCB designs, from the boot ROM and U-Boot through to a booting SMP Linux.
- Designed and integrated Linux kernel drivers for input capture, PWM, quadrature/pulse counters, GPIO and I^2C sensors; upstreamed the liquid-flow sensor driver written for this platform into mainline Linux.
- Developed the secondary Battery Management System for a Redox-Flow battery in Rust/Tokio – pump and valve control, sensor acquisition, threshold- and rule-based safety enforcement, MQTT/Protobuf telemetry and an RPMsg bridge to safety-critical firmware on the Cortex-M4 co-processor.
- Built a QEMU-based emulation of the target platform (including a mainline-submitted AM64x SoC model) so that application software and kernel drivers can be tested and debugged without physical hardware.
- Built a machine-learning soft-sensor that estimates the per-side state-of-charge from operating signals alone (voltage, current, flow, temperature, pump data), replacing ORP probes that exist only in the test rigs – gradient-boosted trees over 47 engineered features, trained on a 2.5M-row corpus assembled from four heterogeneous sources, reaching a held-out mean absolute error of 8% SoC. Wrote the pure-Rust tree-ensemble inference engine that runs the model on the embedded target itself.
- Wrote a high-throughput virtual screening pipeline for redox-active electrolyte candidates: SMILES validation, conformer generation, semi-empirical refinement with GFN2-xTB/CREST (DFTB+ for periodic systems) as a fast filter, and DFT with Psi4 for the accurate analysis of the remaining candidates, with the data layer on Spark/Databricks. This lets the chemists rank and prioritise new molecules computationally before anything is synthesised.

Continental Automotive GmbH

Babenhausen, Germany

Software System Integrator/Embedded Linux Developer — *Jul. 2018 - Jan. 2025*

- Integrated Hardware-Vendor-provided Board Support Packages into Instrument Cluster Products based on Renesas Multicore ARMv7/v8 targets running Greenhills INTEGRITY OS.
- Designed and deployed Build Systems for various Instrument Cluster Products based on CMake.
- Debugged system level errors on Multicore ARMv8/v7 Systems using JTAG with GHS (MULTI) and Lauterbach Tools (Trace32).
- Implemented several RCar3 SoC Device Emulations in QEMU. Deployed machine emulation of customized Renesas RCar3 boards using QEMU https://github.com/wafgo/qemu/tree/rcar3_port.
- Deployed a Yocto based Linux Distribution as a Flash-Frontend for Instrument Clusters running Greenhills INTEGRITY OS.
- Established a core architecture for regulating of outbound DNS traffic in multi-account and multi-VPC environments utilizing AWS Route53 DNS Firewall and FMS. This significantly increased the level of security confidence in the financial sector's segregated environment.
- Developed and Deployed Debugging Tools for GHS based systems to support low level error detection (offline stackdump unwinding on armv8, ARM ETM Setup etc.)

Magna Powertrain

Sailauf, Germany

Embedded Software Engineer — *Jan. 2017 - Jul. 2018*

- Implemented Algorithms for Brushless DC Drive Controls on resource limited Microcontrollers.
- Ported FreeRTOS to the NXP S12Z CPU Architecture.
- Designed a Software Architecture to support the development of MCU agnostic drive control algorithms including kbuild style buildsystem based on GNU make.

Rohde & Schwarz

Mainhausen, Germany

Embedded Software Engineer — *Jan. 2009 - Dec. 2016*

- Implemented several Linux Kernel Drivers for customized Freescale IMX6 based Boards used in Rohde & Schwarz Test and Measurement Devices.
- Developed Bringup Software for Freescale IMX6 and Renesas SH2A Boards.
- Developed a SPI to Ethernet bridge based on an PIC32 MCU with a MCU Firmware and the corresponding Linux Kernel Network driver on the Renesas SH7264 (SH2A).
- Developed and Deployed a Lauterbach Trace32 Debug Script Infrastructure to support early boot debugging.
- Implemented an USB Device Interface based on the TMC (Test and Measurement Standard) which is used in several Rohde & Schwarz Measurement devices for configuration and data retrieval over USB.
- Implemented many Application Level Modules (including corresponding Unit Tests) for several Test and Measurement Devices (LCR/Multimeter/H-Bridge).